

SAN JOSE SANTAMARIA INTL, Costa Rica

WMO: 787620

Lat: 9.9892N

Lon: 84.2183W

Elev: 2943

StdP: 13.20

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	62.4	63.8	48.6	56.6	75.0	51.4	62.8	74.8	17.9	77.1	16.2	76.7	6.6	60	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.9	87.5	69.1	86.1	68.7	84.6	68.6	75.4	80.1	74.5	79.6	73.6	79.1	11.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	140.6	77.7	73.1	137.5	77.2	71.8	131.4	76.1	41.3	80.0	40.2	79.8	39.4	78.4	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	21.2	19.3	DB	59.0	90.7	1.5	2.6	57.9	92.5	57.0	94.0	56.2	95.4	55.1	97.3	(4)
(5)			WB	53.2	77.9	1.6	2.9	52.1	80.0	51.2	81.7	50.3	83.3	49.1	85.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.2	73.9	74.7	75.7	76.2	75.0	74.3	74.4	74.1	73.2	72.8	72.8	73.6	(6)	
	DBStd	2.16	1.76	1.67	1.79	1.86	2.22	1.92	2.01	1.99	1.84	1.99	1.94	1.85	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	8843	742	691	796	787	775	728	757	747	696	706	685	733	(10)	
	CDD65	3368	277	271	331	337	310	278	292	282	246	241	235	268	(11)	
	CDH74	15643	1386	1521	2090	2052	1443	1131	1194	1161	902	784	815	1164	(12)	
	CDH80	3052	224	340	612	622	307	166	183	180	115	79	76	148	(13)	
(14) Wind	WSAvg	9.0	13.3	13.9	12.7	9.8	7.3	6.6	7.8	7.0	5.8	5.6	7.4	10.7	(14)	
(15) Precipitation	PrecAvg	76.9	0.2	0.6	0.7	3.1	10.5	10.2	7.2	10.3	13.5	13.2	5.9	1.4	(15)	
	PrecMax	105.0	1.5	3.1	5.2	13.3	18.8	21.2	16.5	19.6	22.2	21.8	17.9	5.9	(16)	
	PrecMin	50.6	0.0	0.0	0.0	0.1	1.4	0.0	1.7	3.0	5.1	7.1	0.0	0.0	(17)	
	PrecStd	11.9	0.4	0.8	1.0	2.8	3.2	3.5	3.5	4.2	3.9	3.8	3.3	1.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.9	87.6	89.4	89.6	87.8	86.0	85.9	86.1	85.6	84.5	84.2	84.5	(19)	
		MCWB	68.8	67.9	69.3	69.8	70.5	71.4	70.9	71.0	71.0	71.3	70.0	68.6	(20)	
	2%	DB	84.0	85.7	87.1	87.4	85.4	83.9	84.0	84.0	82.7	82.3	82.3	82.8	(21)	
		MCWB	67.0	67.1	68.1	68.8	70.0	71.0	70.2	70.4	71.2	71.2	69.6	67.6	(22)	
	5%	DB	82.5	84.0	85.7	85.8	83.7	82.1	82.3	82.2	80.9	80.4	80.6	82.0	(23)	
		MCWB	66.7	66.4	67.4	68.3	70.5	71.2	69.9	70.3	71.7	71.4	69.6	67.4	(24)	
	10%	DB	80.9	82.3	83.9	84.0	81.6	80.4	80.5	80.4	79.1	78.7	78.8	80.4	(25)	
		MCWB	66.2	65.8	67.0	68.5	70.7	71.3	70.1	70.7	71.5	71.2	69.7	67.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.5	71.7	73.8	74.8	75.6	75.8	75.3	75.6	76.3	75.6	75.1	72.8	(27)	
		MCDB	80.9	80.8	82.8	82.7	81.6	80.2	79.5	79.5	80.0	79.7	79.2	78.9	(28)	
	2%	WB	70.4	69.9	71.5	73.1	74.2	74.5	73.8	74.3	74.8	74.0	73.1	71.3	(29)	
		MCDB	78.8	79.6	81.1	81.5	80.4	79.3	78.9	78.9	79.0	78.6	77.7	77.6	(30)	
	5%	WB	68.7	68.6	70.1	71.8	72.9	73.2	72.6	72.9	73.4	72.8	71.8	70.1	(31)	
		MCDB	78.4	79.3	79.4	79.9	79.2	78.5	78.0	78.2	78.1	77.7	76.8	76.9	(32)	
	10%	WB	67.5	67.4	68.8	70.6	71.8	72.2	71.4	71.8	72.1	71.6	70.7	69.0	(33)	
		MCDB	77.9	78.4	78.9	78.6	78.1	77.5	76.9	77.2	77.1	76.7	75.7	76.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	16.9	18.2	17.9	15.4	14.7	14.6	15.1	15.3	14.0	14.1	14.8	(35)	
		MCDBR	17.7	19.2	20.5	19.8	17.3	16.1	16.1	16.7	16.9	16.2	16.0	16.6	(36)	
	5% WB	MCWBR	8.1	8.3	8.9	8.8	8.5	8.5	8.1	8.8	9.7	9.1	8.9	8.2	(37)	
		MCWBR	17.3	17.9	20.1	18.8	15.9	15.3	15.2	15.8	16.0	15.1	15.2	16.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.347	0.362	0.365	0.395	0.394	0.390	0.409	0.423	0.438	0.373	0.382	0.357	(40)	
		taud	2.556	2.495	2.475	2.398	2.450	2.476	2.413	2.363	2.310	2.545	2.500	2.558	(41)	
	Edn at Noon	Edn at Noon	305	303	302	289	284	282	278	278	277	296	292	300	(42)	
		Edh at Noon	32	35	36	39	36	35	37	40	43	33	34	31	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1501	1715	1787	1663	1426	1371	1306	1384	1442	1298	1205	1335	(44)	
	RadStd	RadStd	107	102	103	82	96	66	69	91	89	81	106	117	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page